ARGUMAT





SPIRAL EVO® TH TL

Self-tipping mixer

Contents

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- Focus on dough mixing
- Recommendation
- Introducing the new range
- Operation and usage

Argumentation

- Ergonomics
- Productivity
- Safety
- Dimensions

The PRO’S

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REMINDER

Focus on dough mixing



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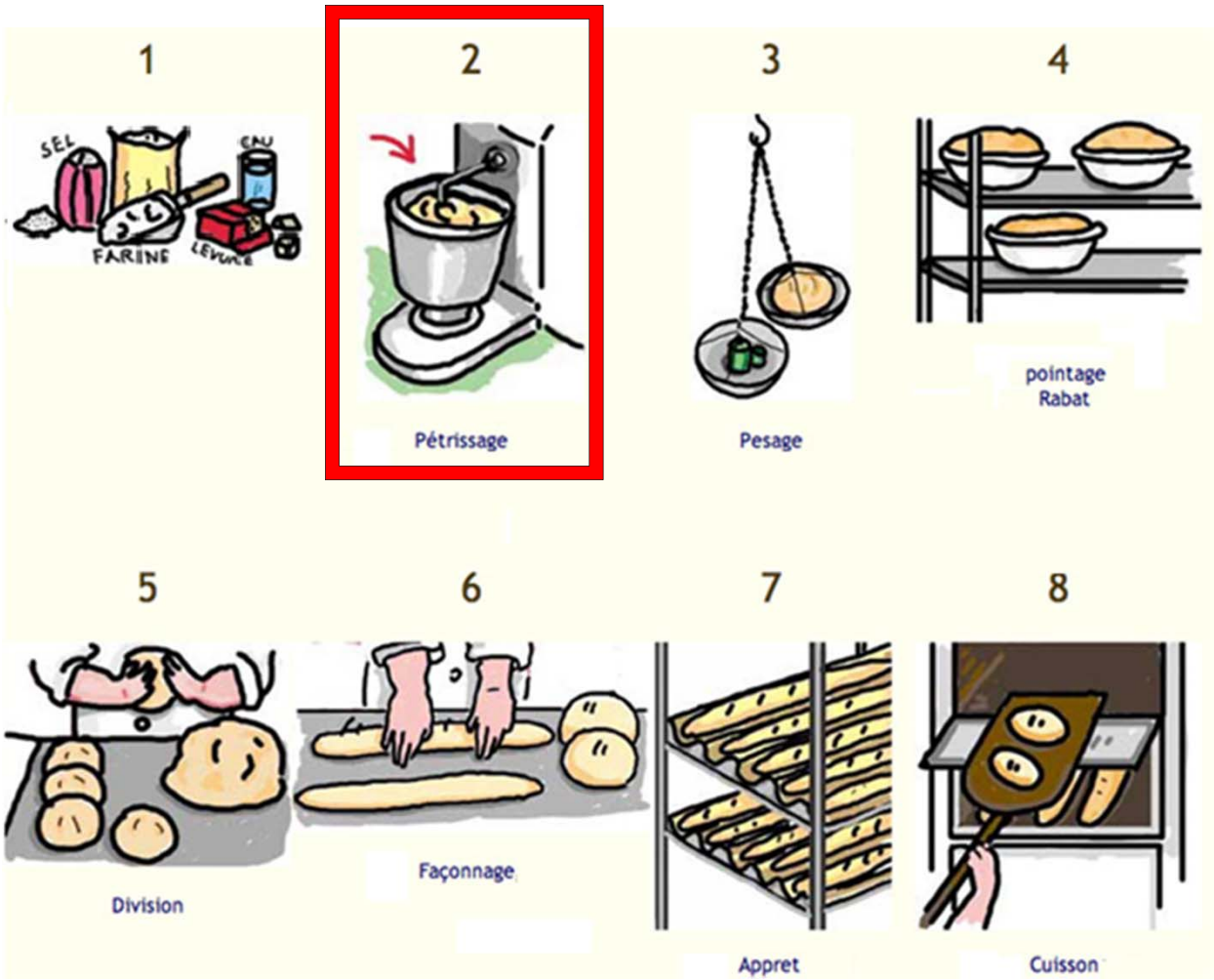
Argumentation

The PROS

Mixing: a key stage

REMINDER

Mixing is the **1st stage** in the bread making process and one that involves ingredients incorporation to create a dough.



BONGARD

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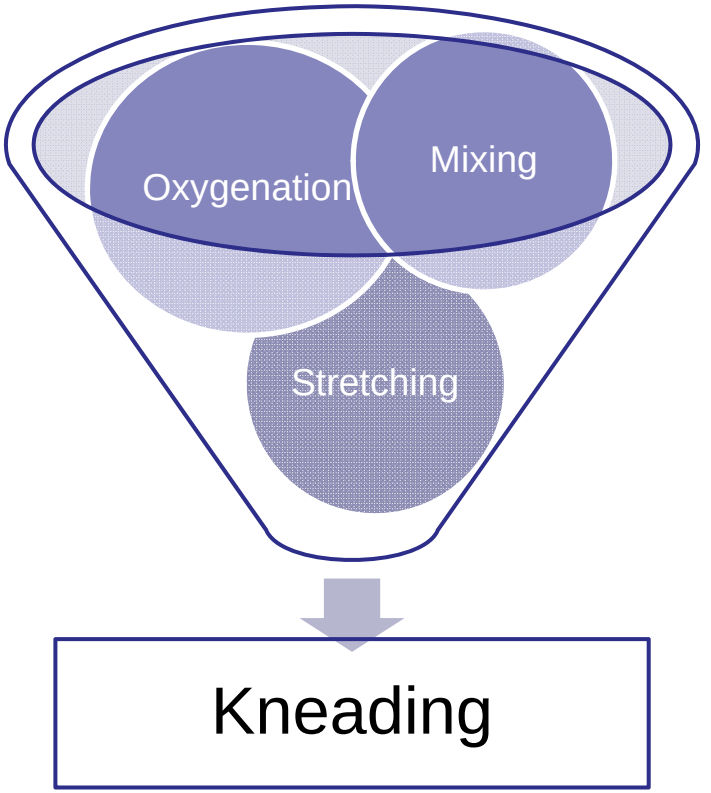
Argumentation

The PROS

What is kneading?

REMINDER

Kneading involves a number of different steps:



1. Mixing: involves **mixing the ingredients** and **forming a dough.**

2. Stretching and oxygenation: helps to strengthen the **gluten strands that give the bread its structure, get air into the mixture** and **increase the volume** of the dough.

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What makes up a good mixer?

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- **Favouring good oxygenation**:







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What makes up a good mixer ?

REMINDER

- **Preventing the dough from getting too warm**

Int. temp. of dough	Consequence
>25°C: dough too warm	• Difficult to work with • Dough is sticky • Fermentation has taken place too quickly
<21°C: dough too cold	• Longer fermentation time • Fermentation has taken too long



Base temp. = Bakery temp. + Flour temp. + Water temp.



dough temp. 22°C ➔ 55°C

dough temp. 25°C

The temperature of the dough at the end of the mixing process is vital where the baker is concerned !

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Recommendation

1) Type of mixer

2) Bowl size

Mixer options for the artisan

REMINDER



Fork Type mixers :
FBL

Traditional French mixer



Spiral Mixer :
SPIRAL

Growing exponentially and available in different versions (A, TH, TL)



Twin arm mixer:
Artofex-type

Not very widely used and not featured in the Bongard range

A comparison of the different technologies

REMINDER



Oblique axis mixer :
FBL

- **Ease** of adjustment
- **Poor pouring range**
- **Bulky**



Diving arm mixer:
Artofex-type

- **Reputation** among professionals
- **Minimal pouring capacity**
- **Slow kneading**
- **Not very well suited** to stronger modern flours or more rapid kneading



Spiral kneading machine:
SPIRAL EVO

- **Rapid kneading**
- **Very broad pouring range**
- **Dough gets warm**

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Recommend a spiral mixer when…

REMINDER



- … the customer is looking for quick mixing



- They want a machine that will allow them to mix different quantities and types of dough



- The customer is short of space

➔ The fast, versatile and compact spiral mixer is compatible with all types of dough and guarantees excellent results.

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Recommend a self-tipping spiral mixer machine (TL) on a work-table when…



- The customer wishes to improve their workplace ergonomics (reducing **M**usculoskeletal **D**isorders)



- The customer adopts a traditional working method involving a work bench



- The customer wishes to increase the productivity of their mixer

➔ Using a TL self-tipping mixer combines ergonomics with productivity, both of which are key factors for a baker!

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The PRO'S

Recommend self-tipping spiral mixer (TH) along with a bread plant



- The customer is using an automatic bread plant and needs to transfer their dough from the mixer into a hopper



- The customer wishes to improve their workplace ergonomics (reducing **M**usculoskeletal **D**isorders)



➔ Enjoy all the benefits of a Spiral EVO mixer (speed, versatility and compactness) whilst also improving ergonomics.

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Recommending a Spiral EVO mixer size:

What does the number referring to the size of the Spiral actually mean?

- Spiral **110**
- Spiral **150**
- ...
- Spiral **300**

Nominal bowl capacity (L)



EXAMPLE	Spiral 150
Max. quantity of flour	62.5 kg
Max. water pouring	37.5 L
Max. quantity of dough	100 kg
Max. bowl capacity	**150 L**

Recommending a Spiral EVO Mixer size:

REMINDER

Replacement

- Recommendation should be based on the customer's **experience** and **development potential**

Creation

- **Organisation** of workflow within the bakery
- **Variety** of doughs
- Max. **quantity** of dough to be produced
- **Oven** capacity
- **Traditional** or **Paneotrad**-based bread making

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The PRO'S

Recommending a Spiral EVO TH or TL mixer size:

Maximum quantity of dough expected per batch :

Min. to max. dough quantity	Recommendation
3.5 kg - 80 kg	Spiral Evo 110
4 kg - 100 kg	Spiral Evo 150
4 kg - 130 kg	Spiral Evo 200
4.5 kg - 200 kg	Spiral Evo 300

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Recommending a Spiral EVO TH or TL mixer size:

Quantity flour used and/or number of units sold:

•Recommendation based on the production of 3 batches

Quantity of flour	Number of units	Recommendation
4,5 – 6,0 tons / month	700 - 900 units/day	Spiral 110 – 150
6,0 – 8,0 tons /month	900 - 1,150 units/day	Spiral 150 – 200
8 – 10 tons /month	1,150 - 1,500 units/day	Spiral 200 - 250
10 – 12 tons /month	1,500 - 1,800 units/day	Spiral 250 - 300

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Recommending a Spiral EVO TH or TL mixer size:

Based on oven capacity:



- In the case of rack ovens, recommend **a mixer size equivalent with dough capacity of 2-baking racks**

8.64: 2-rack ~ 304 baguettes, i.e. 100 kg of dough
➔ **Spiral 150**

- For other types of oven, **ideally** recommend **completing two full oven batches** or at least a little more than one batch **(+10%)**



Orion 802/4.163: One full oven batch ~ 96 baguettes (64 kg)
➔ +10% (71 kg) **Spiral 110**

Min. to max. dough quantity	Recommendation
3.5 - 80 kg	Spiral Evo 110
4 kg - 100 kg	Spiral Evo 150
4 kg - 130 kg	Spiral Evo 200
4.5 kg - 200 kg	Spiral Evo 300

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Introducing the new range

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The PRO'S



Introducing the range

SPIRAL EVO TH

- Spiral EVO 200
- Spiral EVO 300

- Tip to **right** or **left**

- **With** or **without** a chute for tipping into DVP or DVM volumetric dough dividers

SPIRAL EVO TL

- Spiral EVO 110 NEW
- Spiral EVO 150 NEW
- Spiral EVO 200
- Spiral EVO 300

- Tip to **right** or **left**

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Introducing the TL range

	Spiral EvO TL 110	Spiral EvO TL 150	Spiral EvO TL 200	Spiral EvO TL 300
Bowl				
Fixed	■	■	■	■
Controls				
Electronical (only)	■	■	■	■
Manual mode 'Paneotrad'	■	■	■	■
Temperature sensor	€	€	€	€
Cover type				
Transparent lid	■	■	■	■
Stainless steel grid only	□	□	□	□
Bowl				
Stainless steel bowl	■	■	■	■
Power supply				
3N~ 400 V	■	■	■	■

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The PRO'S

Introducing the TH range

	Spiral EvO TH 200	Spiral EvO TH 300
Bowl		
Fixed	■	■
Controls		
Electronical (only)	■	■
Manual mode 'Paneotrad'	■	■
Temperature sensor	€	€
Cover type		
Transparent lid	■	■
Stainless steel grid only	□	□
Bowl		
Stainless steel bowl	■	■
Accessorie		
Dough chute	€	€
Power supply		
3N~ 400 V	■	■

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The PRO'S

New mixer features Spiral Evo TH and TL:

- Optional **infrared temperature sensor**





Continuously measures the surface temperature of the dough





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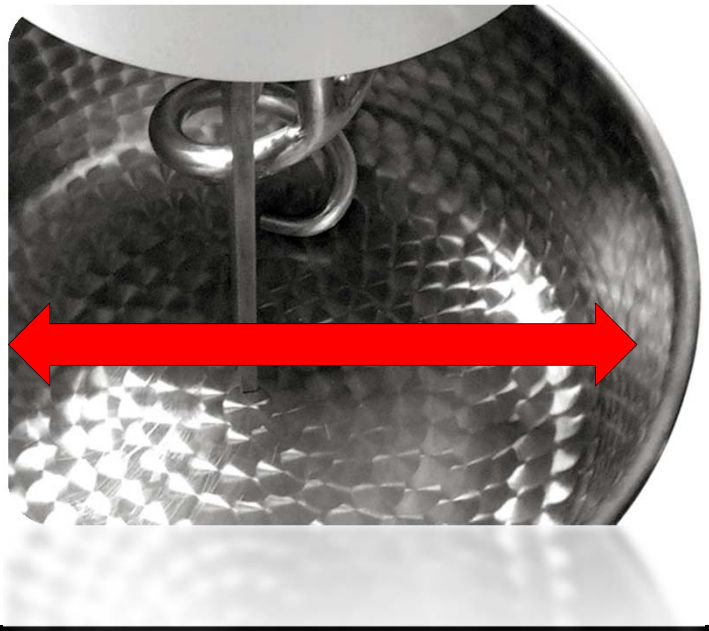
The PRO'S

New kneading machine features Spiral Evo TH and TL:

- New dimensions:

Increased internal bowl diameter

Bowl diameter (mm)		
Models	*Spiral*	*Spiral Evo*
Spiral 110	730	**741**
Spiral 150	730	**741**
Spiral 200	830	**847**
Spiral 300	930	**948**



Introducing the SPIRAL EVO TH and TL range

- Greater dough mixing range

- Previous range	- New range

General features					
Models		*Spiral 110*	*Spiral 150*	*Spiral 200*	*Spiral 300*
Min. to max. dough capacity (kg)		6.4 - 80	5.5 - 100	5.5 - 120	8 - 200
Dough capacity (kg)		**3.5** - 80	**4** - 100	**4 - 130**	**4.5** - 200
Min. to max. rotation speed (rpm)	Bowl	7.5 - 15		12	
	Bowl	7.5 - 15		**10 - 20**	
Min. to max. rotation speed (rpm)	spiral	100 – 200		100 – 210	104 – 208
	spiral	**103 – 207**			
Motor power		4.55 kW		6.65 kW	8.15 kW
Motor power		4.55 kW		**6.8** kW	**8.4** kW

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The PRO'S

New mixer features Spiral Evo TH and TL:

- New PMMA plastic bowl protection guard that is 30% thicker prior to thermoforming



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The PRO'S

New mixer features Spiral Evo TH and TL:

- New control functionalities (including a special Paneotrad manual mode)



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The PRO'S

New mixer features Spiral Evo TH and TL:

- Optional Grid type **safety guard** beneath the compulsory closed safety guard



Mixer available with Grid type safety guard only
but then without a CE certificate

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Operation and use



Mixer components

REMINDER

The mixing process is based largely on the perfect combination of the following factors:

- **The bowl :**



- **The spiral:**



- **The dough breaker bar :**



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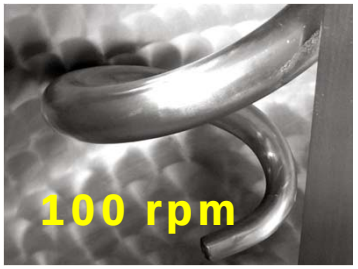
The PRO'S

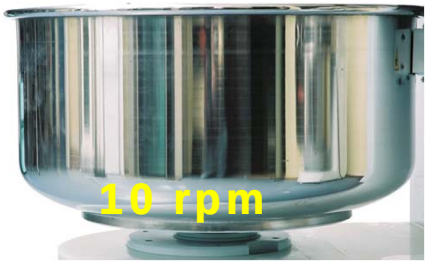
How a spiral mixer works

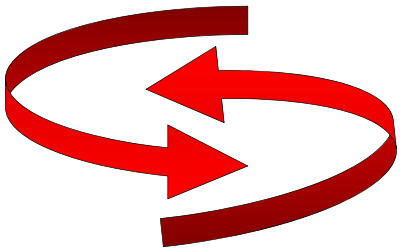
REMINDER

Each mixer comes with two rotation speeds:

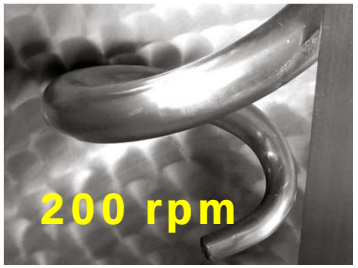
- the **1st speed setting** is used for the **ingredient incorporation** stage (**slow rotation** speed).

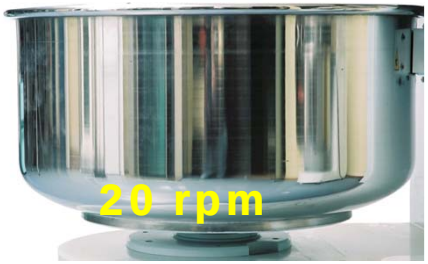






- The **2nd speed setting** is used for **stretching** and **oxygenating** the dough (**fast rotation** speed).





The operating mechanism

- Difference in operation with and without the Grid type safety guard

➢ Single closed cover
or Grid type guard only (for export outside of the EC)

- The mixer stops when the user lifts the cover or grid type guard

- The user can switch to the 2nd mixing speed setting after 2 seconds.

➢ Closed cover + Grid type guard operation

- The user can lift the closed cover without the mixer stopping after the 2-minute ingredients incorporation stage. (minimum required by the EC standard)

- The technician can, if need be, reduce this setting to 2 seconds

Electronic control (only):

Description of control panel

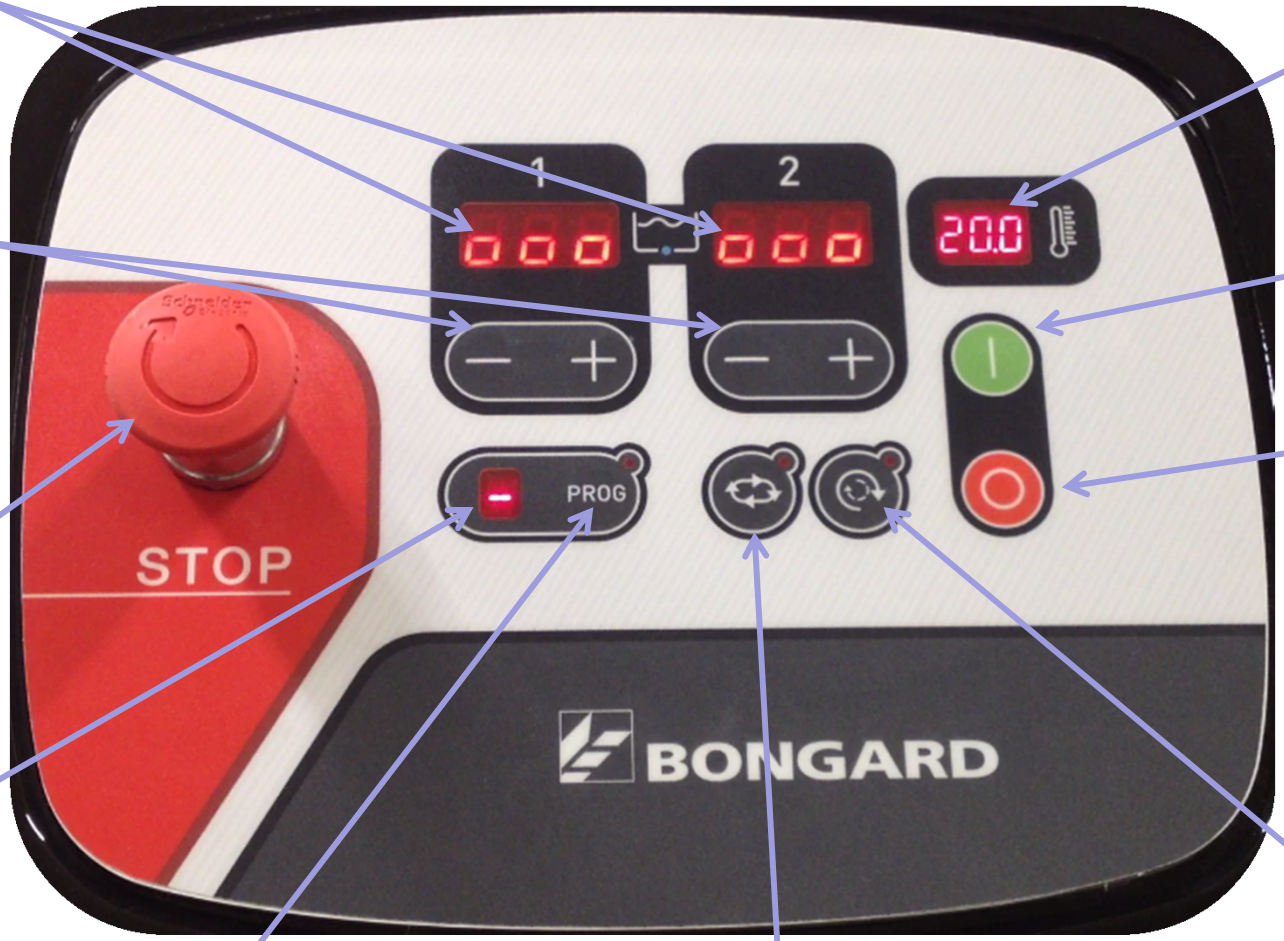
Timer for speed settings 1 and 2

Button for increasing/decreasing mixing time

Emergency button

Active mode displayed: automatic ‘0 to 9’ or manual ‘H’

Programme selection button



Button for activating Manual (H) or Automatic (1, 2, etc.) modes

Mixing temperature display

Start button

Stop button

Button:
- Start of bowl rotation with bowl protection guard open
- Reversal of direction of rotation during the mixing stage*

**dual-motor kneading machines only*

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Controlling the mixer electronically:
Advanced operation

Options:

➔Special 'Paneotrad' manual mode: new feature

➔Stopping the ongoing mixing process based on the temp. (with sensor option only)

Controlling the self-tipping base electromechanically:



Lowering the mixer

Rotating the bowl

Lifting the mixer

Emergency button

Action button

PRESS AND HOLD

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Controlling the mixer electronically:

Advanced operation

Options:

➔Safe working methods: the baker is required to have both hands on the controls in order to activate the hydraulic self-tipping arm, meaning that the tank cannot move if the baker is beneath the mixer

➔Ergonomic use : only 3 action buttons

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Ergonomics



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The PRO'S

Better ergonomics for the baker

Puts less strain on the baker's back :

Transfer:
- to a table
- to a volumetric divider

Advantages:
- Easy removal of dough from mixer bowl
- Direct transfer of the dough **with no manual intervention,** without having to put your hands into the mixer or haul the dough into a divider



Sinmag mixer with low positioned bowl

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Ergonomic controls

- **Intuitive controls** from the very first use





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Control panel comparison

COMPETITION

Non-intuitive VMI controls:

➔ More complex controls with **50% more buttons** than our control panel

➔ Requires a user manual: **available only on CD**





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Control panel comparison

COMPETITION

SINMAG controls:

➔ Controls include many buttons in English

➔ Requires a user manual that exists only **in English**…

➔ **No option of saving** programmes





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PRODUCTIVITY



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Increases the productivity of the mixer

Improved mixer productivity:



The baker can free up their mixer by emptying its content onto a work table, allowing them to start the next batch

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SAFETY

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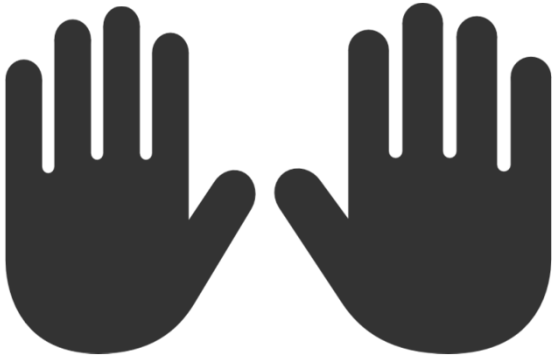
The PRO'S

Safety

Safe working methods:

The baker is required to have both hands on the controls in order to activate the self-tipping hydraulic arm

→ The self-tipping arm cannot operate whilst the baker is beneath the bowl





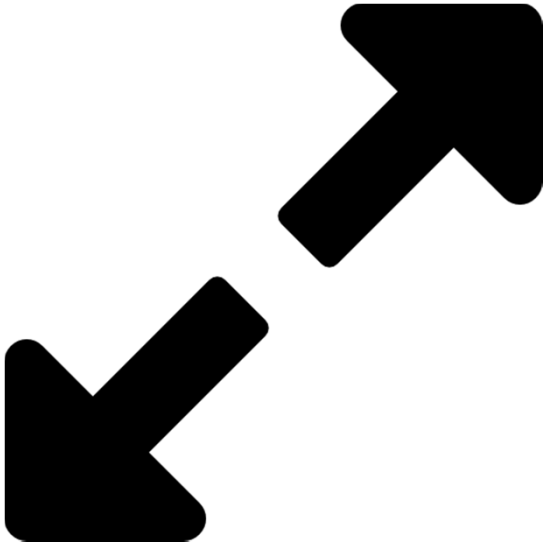
PRESS AND HOLD BUTTON

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DIMENSIONS OF THE RANGE

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The PRO'S

New extension of the range: 110 and 150

The advantages of using a smaller mixer :

- **Less deterioration in the quality of the dough**

The smaller the mixer

The less need there is for the baker to handle large volumes of dough

The less time it takes to cut and shape the dough

The less the quality of the dough will deteriorate as a result

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New extensions of the range: 110 and 150

The advantages of using a mixer :

- **Obtaining a workable dough**

The smaller the kneading machine

The less the dough is exposed to room temperature

The less the yeast grows

The more workable the dough becomes

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New dimensions of the range: 110 and 150

The advantages of using a smaller mixer :

- **Avoiding the risk of blistering during the controlled fermentation stage**

The smaller the mixer

The less dough I have to work

The less time the dough in the mixer will develop

The risk of blistering is getting lesser

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New extensions of the range: 110 and 150

The advantages of using a smaller mixer:

- **Favouring good dough development in the oven (as part of the Paneotrad process)**

The smaller the mixer

The smaller the volume of dough that is handled

The less time it takes the baker to fold the dough

The quicker the dough will be retarded

Limits excess fermentation

Favours good development in the oven

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THE PRO'S



THE PRO'S

CUSTOMER GROUP	ARTISAN	SUPERMARKET	CHAIN stores
Cooking capacity:			
Ergonomics	V	VV	VV
Productivity	V	VV	VV
User safety	V	VV	VV
Dimensions	VV	V	V